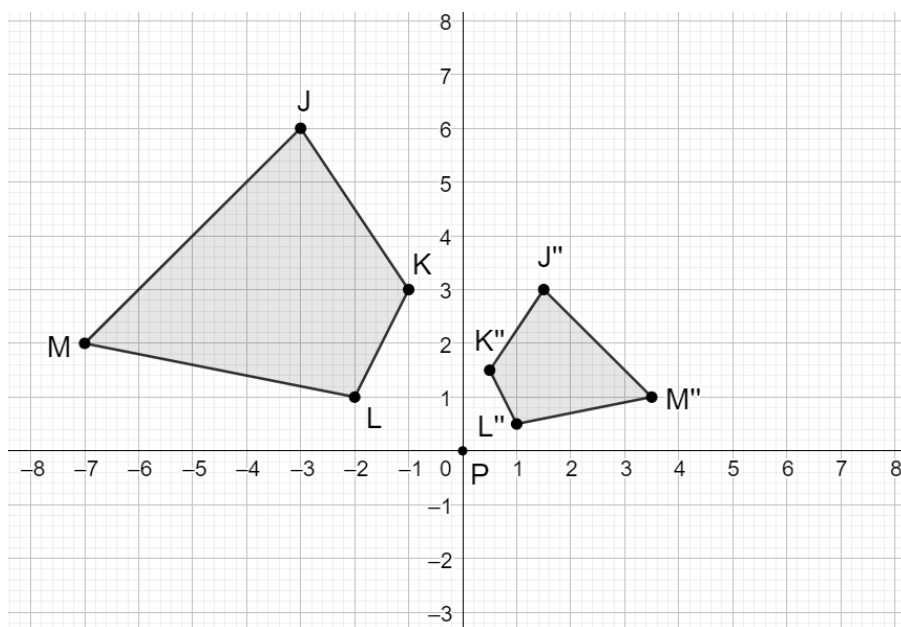


Geometry
Unit 6
Lesson 6 Homework

Name Answer Key
Date _____
Period _____

1. When a sequence of transformations results in both angles and sides of the image that are congruent to corresponding angles and sides of the preimage, the sequence ☐ does ☒ does not include a dilation. If it ☒ does ☐ does not include a dilation, the corresponding ☒ angles ☐ sides are congruent while the corresponding ☐ angles ☒ sides are proportional.



2. Write a sequence of transformations for preimage $JKLM$ that would show whether the figures are similar.

Reflection over y -axis and dilation with scale factor $\frac{1}{2}$

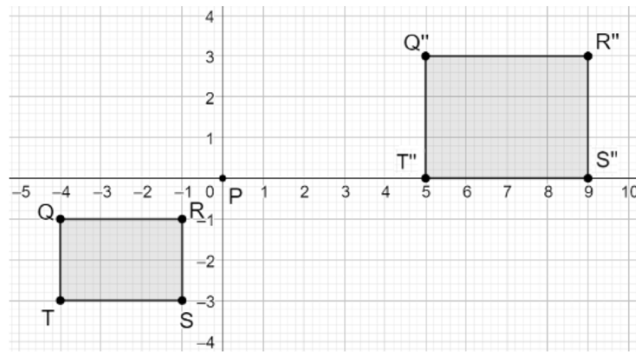
$$(x, y) \rightarrow (-x, y) \text{ and } (x, y) \rightarrow \left(\frac{1}{2}x, \frac{1}{2}y\right)$$

3. Is $J''K''L''M''$ similar to the preimage?

☒ Yes
☐ No

4. Explain the reason for your answer for #3.

Angles are congruent and sides are proportional.



5. Write a sequence of transformations for preimage $QRST$ that would show whether the figures are similar.

Figures are not similar.

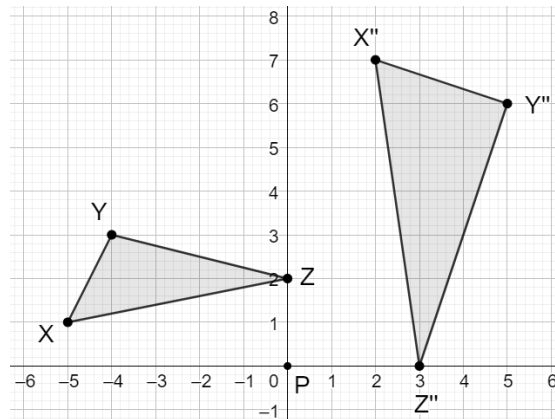
6. Is $Q''R''S''T''$ similar to the preimage?

☐ Yes

☒ No

7. Explain the reason for your answer for #6.

The sides are not proportional.



8. Write a sequence of transformations for preimage $\triangle XYZ$ that would show whether the figures are similar.

Figures are not similar.

9. Is $\triangle X''Y''Z''$ similar to the preimage?

☐ Yes

☒ No

10. Explain the reason for your answer for #9.

The sides are not proportional.